

Utkarsh Somvanshi

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Summary

Second-year Computer Science student specializing in AI, with hands-on experience across machine learning, computer vision, and full-stack development. Built and deployed multiple end-to-end AI/ML systems—from real-time object detection to predictive analytics—combining strong statistical fundamentals with practical engineering using Python, FastAPI, and modern data tools.

Technical Skills

Languages: Python, SQL, HTML & CSS

Machine Learning & Statistical Modeling: Scikit-learn, XGBoost, Regression & Classification, Ensemble Methods, Time-Series Analysis (ARIMA, SARIMA), Predictive Modeling

Computer Vision: YOLO, Real-Time Object Detection

Backend & APIs: FastAPI, REST APIs, PostgreSQL, WebSocket, SQLite

Frontend: React, JavaScript, Tailwind CSS

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI, Excel

Tools: Git, GitHub

Experience

ArcelorMittal Nippon Steel India (AMNS)

May 2026 – Present

AI & Computer Vision Engineer Intern

On-site

- Built SafeVision AI, an in-house computer vision safety monitoring platform across 10+ AI cameras, currently in final testing to replace AMNS's third-party DARSA vendor system — eliminating recurring licensing costs
- Trained 4+ YOLO-based object detection models for PPE compliance, achieving up to 97% mAP50 and 95%+ average precision/recall across helmet, safety vest, and restricted-zone violation classes
- Engineered real-time detection across 5+ violation types (missing helmet, missing safety jacket, work-at-height breaches, crowd density) with automated alert generation
- Developed a full-stack platform covering dashboards, alerts, compliance tracking, live camera monitoring, and admin settings, with WebSocket-based live updates
- Designed broader system functionality and admin control than the incumbent vendor solution, which offered only login-based dashboard access with no configurability

Unified Mentor

Jan 2026 – Apr 2026

Data Analytics Intern

Remote

- Delivered 5 end-to-end data analytics projects spanning retail, finance, inventory, and customer support domains, each following a structured 8–10 phase analytical workflow (EDA, feature engineering, SQL-style business queries, ML modeling)
- Built a stock price prediction model on 20 years of TCS trading data (4,463 records), achieving $R^2=0.996$ using Random Forest on next-day return forecasting
- Conducted large-scale inventory analysis across 6 datasets (2.3M+ purchase records) for a beverage distribution business, applying ABC classification and turnover analysis to identify working-capital inefficiencies
- Performed sentiment and market intelligence analysis on 9,600+ Google Play Store apps and 64,000+ user reviews, surfacing actionable product and monetization insights
- Built and compared 4 classification/regression models (Logistic Regression, Random Forest, Gradient Boosting, XGBoost) across projects, with feature importance analysis driving business recommendations

Projects

SafeVision AI – Industrial Safety Compliance Platform | *YOLO, FastAPI, PostgreSQL, React, WebSocket*

- Built an in-house computer vision safety monitoring platform across 10+ AI cameras, currently in final testing to replace a third-party vendor system at AMNS
- Trained 4+ YOLO-based detection models for PPE compliance, achieving up to 97% mAP50 and 95%+ average precision/recall, with real-time dashboards and automated alerts

MoneyMind AI – AI-Powered Personal Finance Advisor | *React, FastAPI, Groq (LLaMA 3.3 70B)*

- Built a full-stack platform offering CA-level financial guidance, including a Money Health Score, Tax Wizard (Old vs. New regime), and Portfolio X-Ray (XIRR, fund overlap, rebalancing plans)
- Integrated an LLM-powered chat advisor (Groq LLaMA 3.3 70B) and PDF parsing to auto-extract data from Form 16 and CAMS/KFintech statements

Mess Demand & Food Waste Management System | *Python, FastAPI, React, Scikit-learn, XGBoost*

- Built an ML pipeline solving a real campus problem — predicting university mess demand (Low/Medium/High) to reduce food over-preparation and shortages, with 6 models benchmarked and Random Forest selected as best
- Implemented automated retraining, model versioning, data drift detection (PSI, Chi-Square, Mean Shift), and dual dashboards (React + Streamlit) with food waste cost estimation

FinAI – AI-Powered Mutual Fund Portfolio Rebalancing Platform | *Python, FastAPI, React, Supabase*

- Built a platform computing a real-time Rebalance Score (0–100) using a weighted 5-factor model and a Bayesian decision engine for portfolio classification
- Implemented a fund recommendation engine with STCG/LTCG tax-zone detection and an interactive what-if allocation simulator

Education

Vijaybhoomi University

Sep 2024 – Sep 2028

B.Tech – Computer Science Engineering with majors in AI

Current CGPA: 7.8

Certificates

Deloitte Australia – Data Analytics Job Simulation

Forage

Tata – Data Visualisation: Empowering Business with Effective Insights Job Simulation

Forage

Community Experience

Light House Communities

May 2025

NGO Summer Internship Program – Tech Trainer. Taught MS Word, Excel, and email to underprivileged learners (18–30 years old). Conducted hands-on training sessions and improved digital confidence of learners.

Sports Committee

Sep 2024 – Present

Organized sports events, managed participants, and coordinated team activities for campus events.

Language

English | Marathi | Hindi